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Research Article

Performance of a Small-Scale Haber Process: A Techno-Economic Analysis

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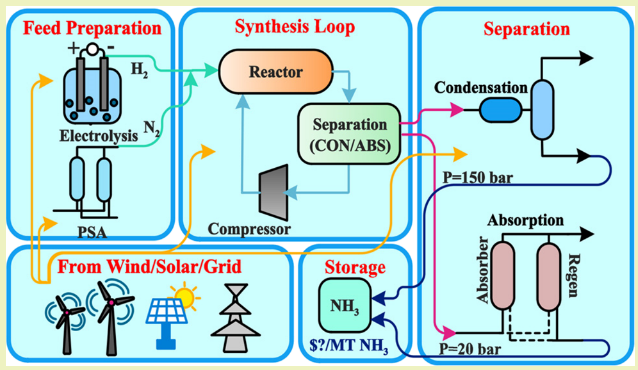

ABSTRACT: In this work, the techno-economic analysis of a 20,000 metric ton (MT) green ammonia production facility is presented. This facility is 30 times smaller than a large-scale conventional process, producing ammonia from totally renewable resources: hydrogen from water electrolysis and nitrogen from pressure swing adsorption. Two different configurations of the Haber−Bosch (HB) process are investigated: high-pressure reaction-condensation (RXN-CON) and low-pressure reaction-absorption (RXN-ABS). Process simulation was implemented using ASPEN Plus, where the reactor and absorber columns were designed as a custom model. The results obtained were then used to estimate the total capital and operating costs. The high-pressure processing improves the single-pass conversion and loop efficiency but relies on costly compression, whereas the low-pressure processing is more favorable for both capital and operating costs. The performance analysis of the HB process indicates that the operating pressure affects ammonia production costs. The levelized cost of ammonia (LCOA) from our small-scale Haber process was found to be about twice more expensive than the conventional commodity ammonia prices. Our sensitivity analysis suggests that inherently safer low-pressure RXN-ABS can be utilized for thermochemical energy storage of renewable resources—for scenarios that numerous small ammonia plants can be implemented in areas with local ammonia demand, with access to excess renewable electricity at the time of high penetration of renewable resources. Under such conditions, the LCOA from this plant can be comparable with the ammonia commodity prices. When the revenue from selling oxygen is considered into economics, small-scale all-electric ammonia can be profitable with an after-tax rate of return of 27.50% for RXN-ABS.

KEYWORDS: *ammonia, Haber−Bosch, renewable, sustainability, absorption, distributed manufacturing*

■ INTRODUCTION

Ammonia produced from hydrogen and nitrogen through the well-known Haber−Bosch (HB) process is one of the most important innovations of the 20th century.[1] This process is critical to sustain the food production, as it is an indispensable ingredient of nitrogen fertilizers.[2] Not only a fertilizer, ammonia is recently proposed as an indirect form of hydrogen storage because of its high hydrogen density (17.8 wt %) and the viability of long-term storage and delivery.[3] Extensive efforts have been undertaken to improve and optimize the HB process to enable cheaper production that meets human demands.[4,5] In the current state-of-the-art technology, steam reforming of natural gas is used to produce hydrogen, as air is introduced into the secondary reformer to supply nitrogen. The gas mixture undergoes multiple reaction and separation units to produce a pure hydrogen and nitrogen mixture. This gas mixture is then compressed into a synthesis loop with a HB reactor operating at high temperature (350−520 °C) and high pressure (150−300 bar). The overall energy consumption of the steam methane reforming (SMR)-based process is approximately 28.0−32.6 GJ/ton of ammonia.[4,6] To reduce the costs, ammonia manufacturing usually occurs in large-scale centralized facilities to take advantage of the economy of scale. In the 21st century, however, there is a gradual progress toward on-site ammonia production that works based on renewable resources.[7]

"Green" ammonia produced with hydrogen from water electrolysis and nitrogen from air, which is referred to as "all-electric" process, has been considered as an alternative to the fossil fuel-based ammonia production.[8,9] The stranded wind and solar energy are available in agricultural regions (e.g., Minnesota River Valley) with huge ammonia demand, which makes the small-scale all-electric ammonia projects more

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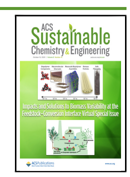

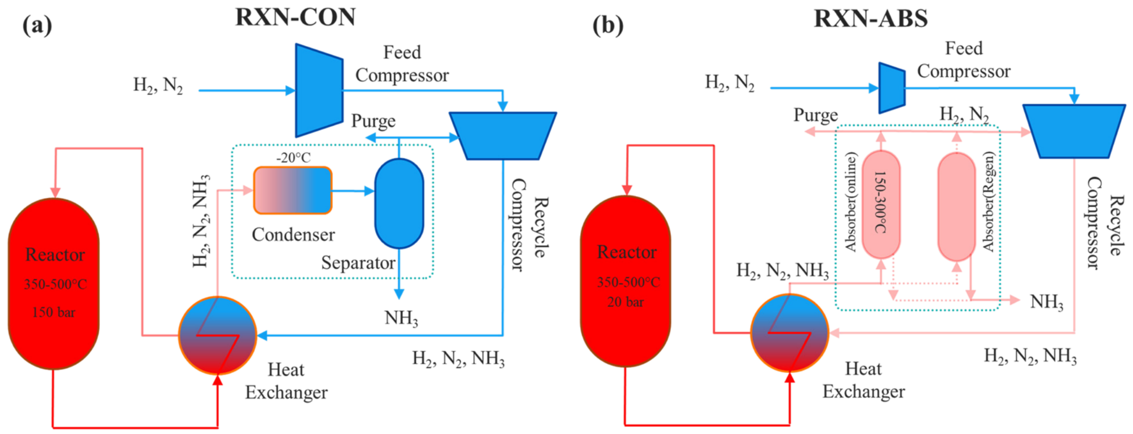

Figure 1. Ammonia synthesis loop block flow diagrams for (**a**) RXN-CON and (**b**) RXN-ABS. RXN-CON represents the synthesis loop in a conventional HB process. In RXN-ABS, the condenser is replaced by absorber columns.

justifiable.[10] According to Lazard's levelized cost of energy (LCOE) analysis, the cost of wind- and solar-generated electricity is comparable or becoming cheaper than the conventional electricity, ranging from \$28 to \$54/MWh.[11] Flexible markets with economically affordable and geographically available energy resources can accelerate the progress toward more sustainable ammonia production. As a result, the idea of installing small, decentralized all-electric ammonia production has received great attention because such facilities can potentially supply fertilizers for sustainable farms while avoiding transportation costs and uncertainties associated with commodity ammonia. However, the major challenge is the fact that the HB process is well optimized for large-scale processing, whereas wind and solar resources can only support small-scale chemical processing. Hence, there exists a disconnect between the size of renewable resources and ammonia production processing at current scales.

In the conventional process, ammonia is produced at high pressure and separated by phase-changing condensation, hereafter referred to as reaction-condensation (RXN-CON). An alternative production route is proposed that produces ammonia at a reduced pressure where ammonia is removed more completely with selective metal halide absorbents at high temperature, hereafter referred to as reaction-absorption (RXN-ABS).[12] Prior literature shows that reactive separation with RXN-ABS provides a pathway to achieve close to 99% conversion.[13,14] Compared to RXN-CON, another major benefit of RXN-ABS is that it allows higher production rates when operated at a lower pressure.[13−16] Lower operating pressure is a significant benefit for distributed manufacturing because it will result in an inherently safer design, as such small plants will be operated by nontechnical operators. While the kinetics of ammonia uptake and release in metal halide absorbents are still under investigation,[17] pressure and temperature swing absorption (PSA/TSA) is believed to be the right approach to operate the absorber column for steady-state production of ammonia with RXN-ABS.[18] Nevertheless, there is a critical need for a detailed performance and techno-economic analysis comparing the levelized cost of ammonia (LCOA), which is defined as the sum of the present value of capital and operating costs over the lifetime of the ammonia plant divided by the total ammonia production, from RXN-ABS and RXN-CON configurations.

Overall, natural gas-based ammonia production is more economically favorable over the renewable-driven production in terms of the LCOA. Morgan et al. provided a detailed economic analysis of a large-scale wind-powered ammonia production with dedicated offshore wind.[19] Their analysis of a 300 ton/day offshore ammonia plant through RXN-CON at 150 bar indicates that the major contribution to the LCOA comes from the installed cost of the offshore wind. The baseline LCOA obtained by Morgan et al. is three times higher than the cost of conventional ammonia at the time of analysis.[19,20] It is important to note that the installed cost of wind has been declining by 70% since 2009,[11] which is in favor of the all-electric ammonia. Sanchez and Martín provided a detailed report on the production of green ammonia from water and air, in a 300 MT/day HB process operating at 170 bar. This report mostly focuses on designing the energy sources (wind and solar) and the reactor layout. The cost of ammonia was found to be ∼\$1600/MT.[21] Palys et al. reported the performance of the ammonia production through RXN-ABS, by optimizing the net present cost of the electrolysis-based ammonia at different production scales.[22] In their work, however, the authors did not focus on the cost of hydrogen and nitrogen production and a detailed cost analysis along with the LCOA is not reported. To date, there is no comparative study reported to compare RXN-CON and RXN-ABS configurations. Overall, it is expected that operating at a reduced pressure improves the economics of RXN-ABS, in terms of both capital and operating costs. None of the prior efforts provide a detailed analysis on the effect of the operating pressure on the performance of HB process.

This work focuses on the techno-economic analysis of small-scale electrolysis-based ammonia production with the aim of comparing RXN-CON and RXN-ABS configurations. A comprehensive process simulation was prepared using Aspen Plus based on the state-of-the-art ammonia production process, to evaluate the effect of the operating pressure and provide the technical information for the cost estimations. A rigorous absorber column for RXN-ABS was also developed using Aspen Custom Modeler (ACM). We used PSA/TSA to operate the metal halide absorber columns integrated with the HB reactor. A comprehensive sensitivity analysis was performed to reveal the effect of the operating pressure and optimize the ammonia production process with each configuration. For the cost analysis, the operating pressure of 150 bar was chosen as the base condition for RXN-CON, which was compared against RXN-ABS operating at 20 bar. In this work, we assumed that the major portion of the electricity

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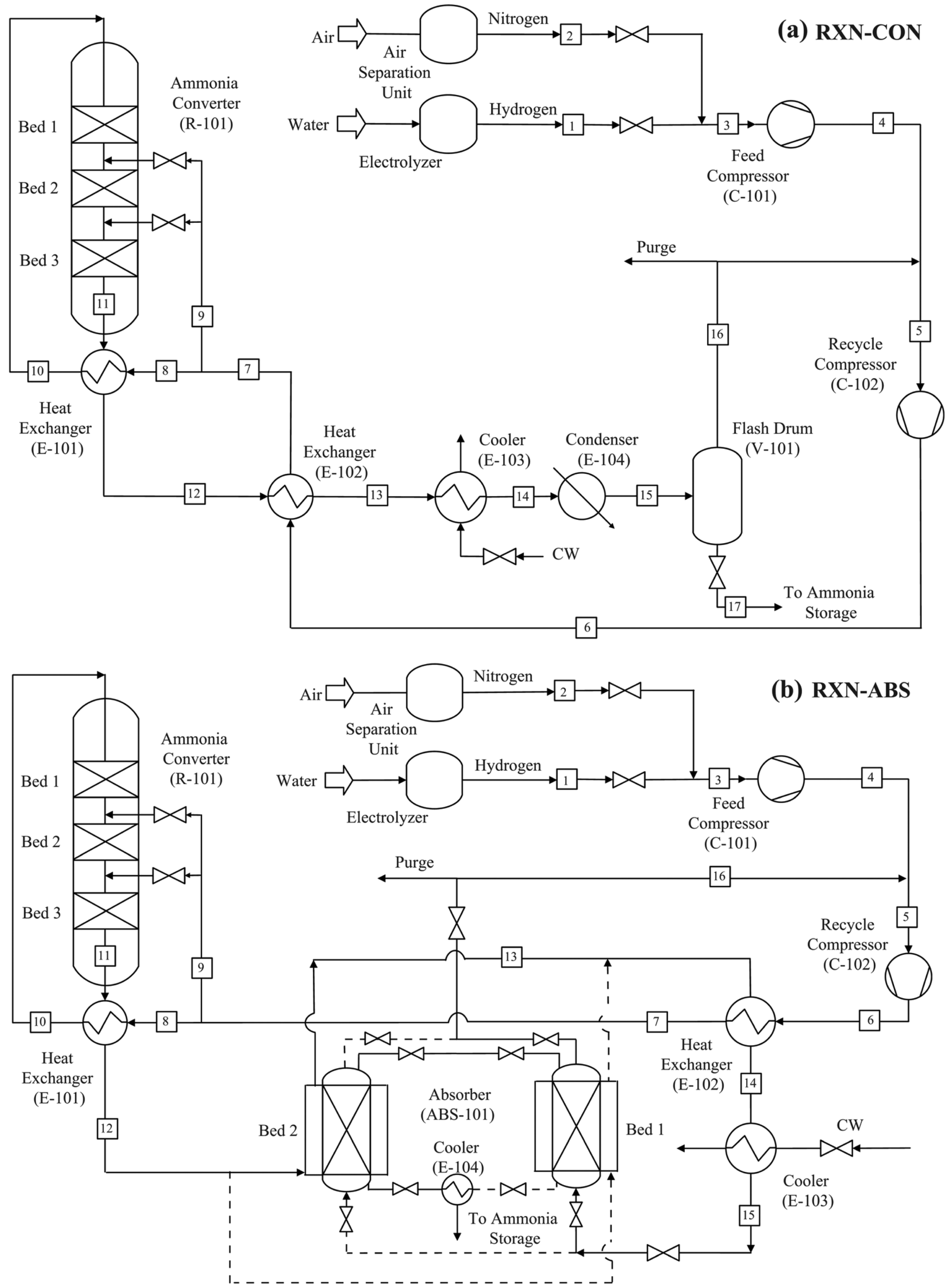

Figure 2. Schematic diagram of ammonia production for (a) RXN-CON and (b) RXN-ABS. The dashed streamline in the RXN-ABS process indicates the cyclic nature of the absorber columns.

will be obtained through a power purchase agreement with a supplier of "green energy." By green energy, we mean electricity derived as much as possible from wind, as the electricity from wind farms is sold at a fixed price over long-period contracts. At the times of intermittency, we assumed to purchase the electricity from the grid at a fixed average cost. Thereafter, the comparative techno-economic analysis of small-scale electrolysis-based RXN-CON (small-E-RXN-CON) and RXN-ABS (small-E-RXN-ABS) was carried out in terms of energy consumption, operating cost, and total capital cost. A detailed discussion on scenarios to further reduce the LCOA is presented.

■ METHODOLOGY

Process Description. The process design approach is described in this section. This work mainly focuses on the synthesis loop consisting of an ammonia converter with intercooling by quenching streams, feed and recycle compressors, and a series of heat exchangers.[23] Block flow diagrams of the synthesis loop for the RXN-CON and RXN-ABS processes are shown in Figure 1. As displayed, they have some features in common, including the ammonia converter, heat exchangers, and compressors. The major difference between RXN-CON and RXN-ABS configurations is the separation unit. In the former (see Figure 1a), a refrigeration system and a flash drum are required to liquefy ammonia from the gas mixture, whereas the latter operates continuously with

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absorber columns. It is important to note that absorber columns operate in transient fed-batch mode. Therefore, at least two columns were required to ensure cyclic steady-state operation, as shown in Figure 1b.

Ammonia Converter. Ammonia is produced from the reaction of nitrogen and hydrogen at a high temperature and a high pressure in the presence of a transition metal heterogeneous catalyst. The reaction is as shown in eq 1:

$$N_2 + 3H_2 \rightleftarrows 2NH_3 \; \Delta H_{298K} = -92.44 \frac{kJ}{mol} \tag{1}$$

This reaction is reversible and exothermic; it is favored by high pressure and low temperature.[6,24] Assuming that N_2 dissociation is the rate-limiting step, all conventional 3d and 4d transition metal catalysts utilized to date show a scaling relation between the transition-state energy required for N_2 dissociation and the adsorption energy of intermediates.[25] Hence, high-temperature reaction is required, which necessitates high-pressure processing to increase the single-pass conversion.[25,26] This is the reason ammonia reforming takes place at elevated temperature (>400 °C) and pressure (>100 bar).[4]

Process flow diagrams for RXN-CON and RXN-ABS configurations are shown in Figure 2. Ammonia synthesis is exothermic; thus, the heat of reaction is released as the reaction proceeds and the gas stream temperature rises along the length of the reactor, which results in a lower equilibrium conversion (see Section 1.1. and Figure S1 in the Supporting Information). Different designs for the ammonia converter were reviewed, and it was concluded that almost the entire ammonia industry utilizes intercooling either by quenching streams (cold shot) or internal heat exchanging.[4] The overall performance of both converters is very similar. In this work, both RXN-CON and RXN-ABS configurations utilize a three-bed adiabatic ammonia converter with a quenching system.

RXN-CON. The process flow diagram for the RXN-CON is shown in Figure 2a. Hydrogen from water electrolysis and nitrogen from air separation are fed to the process (4), compressed by the feed compressor (C-101), mixed with the recycled flow (16), and further compressed by the recycled compressor (C-102) before entering the converter (R-101). Reactants split into a quench stream (9) and a main stream (8). The main stream (8) heats when flowing through the heat exchanger (E-101) and enters Bed 1 where partial conversion is achieved, and the gas mixture further heats up. The product of Bed 1 is cooled and diluted by directly mixing with a portion of the quenching stream (9). The gas mixture then flows to Bed 2. The quenching is necessary to achieve higher ammonia conversion. The gas mixture follows a similar trend before entering Bed 3. To recover the heat of reaction from the reactor effluent (11) and preheat the feed (6), heat exchangers (E-101 and E-102) are designed for heat integration. To separate the ammonia, this process requires a cooler (E-103) and a refrigeration system (here simplified and shown as E-104) to reduce the temperature of the effluent to 40 °C (14) and −20 °C (15), respectively. The simulation of a typical refrigeration cycle was carried out to provide information on heat duty and sizing for comparison in the economics analysis. The liquefied ammonia is then separated in a flash drum (V-101). Liquid ammonia products with a purity of 99.6% are obtained at operating conditions of −20 °C and 150 bar.

RXN-ABS. The process flow diagram for RXN-ABS is shown in Figure 2b. The process outline is similar to RXN-CON, except that the condensation system (E-104 and V-101) is replaced by an absorber column (ABS-101). For proper heat integration, the converter effluent is first directed to one of the absorber column shell side (ABS-101), which is in desorption mode, where the heat of reaction released in the R-101 is used to heat the absorber and regenerate the absorbents by providing the heat of desorption.

■ SIMULATION

Ammonia Reaction. The ammonia synthesis kinetics described by Nielsen et al. is adopted and implemented using a user-defined Fortran subroutine in ACM.[26,27] (see details in the Supporting Information Section 1.1).

Process Simulation. Material and energy balances in RXN-CON and RXN-ABS were carried out in ASPEN Plus for the production of 20,000 MT ammonia annually. This size was selected to compare the finding with other commercial small-scale ammonia plants, e.g., Proton Venture NFUEL.[28] This software has been widely used for the design and simulation of chemical processes, including ammonia.[29−31] The Redlich−Kwong−Soave (RKS)−Boston−Mathias (BM) package was selected and used in this work.[32] The design specifications of process configurations shown in Figure 2 are listed in Table S3 in the Supporting Information Section 1.2. A simplified refrigeration cycle was also developed for RXN-CON for the ammonia liquefaction. The heat exchanger area of a condenser, as well as the compressor power requirements, were obtained from process simulations.

Our immediate goal in process simulation efforts was to understand how the reduced operating pressure could benefit the HB process. We varied the operating pressure and compared the performance of the high-pressure RXN-CON (110−150 bar) with the low-pressure RXN-ABS (20−35 bar). The comparison of two ammonia configurations was carried out in terms of ammonia mole fraction and reactant temperature in a three-catalytic bed ammonia converter. It should be noted that process units were sized and designed to achieve the desired production rate of 20,000 MT/year and stream factor of 0.94, as the operating pressure was varied. We then limited our analysis to RXN-CON operating at 150 bar and RXN-ABS operating at 20 bar, and then compared these configurations with respect to their energy consumption.

Absorption. Since ASPEN Plus does not include a dynamic absorber model, a custom model was developed for the absorber columns using ACM. Mass balances in both the axial coordinate of the bed and the absorbent phase were considered in our model. We assumed an isothermal system due to the minor temperature change (<3 °C) during the ammonia absorption.[33] The transport governing equations for fixed-bed absorption are summarized in Table S7 in the Supporting Information Section 1.3. The mathematical model is solved by the finite difference method using the initial and boundary conditions. The model was then validated against experimental results to accurately predict the breakthrough time. The validated model was used and scaled up to the unit size capable of producing 20,000 MT per year (see Figure S3 in the Supporting Information Section 1.4).

PSA/TSA Cycle. Absorption is a dynamic process and can never reach a steady-state condition. Therefore, three absorber columns were considered for the RXN-ABS process simulation, which enables cyclic steady-state operation for continuous ammonia production. To release the ammonia and operate the absorber in a steady-state condition, cyclic PSA/TSA was considered and simulated in our RXN-ABS process. While one of the beds was in the process and separating ammonia, the other two beds were regenerating (PSA/TSA) and releasing ammonia. More information regarding the heat integration strategy is provided in the Supporting Information Section 1.5. The PSA/TSA conditions proposed for our process is a Skarstrom cycle type consisting of four consecutive steps that is illustrated in Figure 3.[33,34] We assumed that absorber

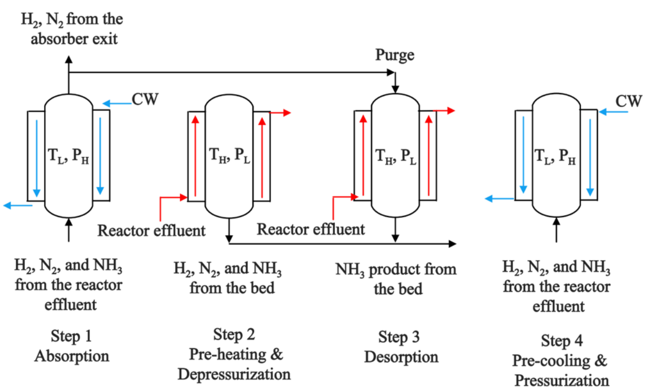

Figure 3. Schematic diagram of the PSA/TSA process proposed for the continuous operation of the absorber column in RXN-ABS. A minimum of three absorbers are needed for the cyclic PSA/TSA operation. (CW: cooling water).

columns were packed with supported metal halides with reproducible cyclic ammonia separation.[18,35] Among absorbents studied to date, $MgBr_2$−Si showed the greatest working capacity between two temperatures: 62 $mg_{NH3}/g_{sorbent}$ at 150 °C and 21 $mg_{NH3}/g_{sorbent}$ at 300 °C.[18] We considered 150 and 300 °C as the absorption (T_L) and desorption (T_H) temperatures, respectively. The absorber was designed to separate ammonia in the process pressure (P_H = 20 bar) and release ammonia at a lower pressure (P_L = 16 bar). The amount of ammonia absorbed in the supported absorbent at P_H and T_L is determined by its apparent capacity (see Figure S4 in the Supporting Information Section 1.4). It was assumed to cycle the absorber columns every 90 min, as below:

Step 1: *absorption* (30 min)—The absorber column is in the absorption mode at high pressure (P_H) and low temperature (T_L). The absorbent removes the ammonia from the reactor effluent. Once the ammonia concentration in the outlet reaches a threshold of 5% of the inlet concentration, the absorber column switches to desorption mode. The exotherms of the absorption were controlled by cooling the absorber with chilling water, as described in the Supporting Information Section 1.5.

Step 2: *preheating and depressurization* (30 min)—The column temperature increases to T_H and the pressure decreases to P_L by opening the bottom valve.

"Step 3: *soaking* (15 min)—As P_L and T_H are reached, the absorber will be kept at these conditions for a more complete ammonia release.

Step 4: *cooling and pressurization* (15 min)—Once the absorbent is fully regenerated, the temperature of the column decreases to T_L using the cooling water through the shell side and the pressure increases to P_H using the reactor effluent to prepare for the adsorption mode.

The order of PSA/TSA cycling of three absorbers is provided in Table 1.

Table 1. Sequence of Operation for the Three-Bed PSA/TSA System[a]

Absorber Column	Steps		
Column 1	step 1	step 2	step 3 step 4
Column 2	step 3 step 4	step 1	step 2
Column 3	step 2	step 3 step 4	step 1

[a](Step 1: feed absorption for 30 min; Step 2: preheating and depressurization for 30 min; Step 3: soaking for 15 min; Step 4: cooling and pressurization for 15 min).

Economics. The simulation results from ASPEN Plus were used to estimate the economics of electrolysis-based RXN-CON and RXN-ABS. Equipment sizes, power requirements, flow rates, and operating conditions were incorporated into our cost analysis to calculate the net present value. The total cost of ammonia production consists of three main subsystems: electrolysis, air separation, and ammonia synthesis loop. We assumed that all-electric ammonia plants will be built in areas where wind-generated electricity is abundant and accessible; hence, only electricity costs are considered in the economic analysis. The capital and operating costs were calculated based on the correlations given by Perry.[36]

For the proposed production capacity, alkaline electrolyzers and PSA were selected as the preferred method for supplying hydrogen and nitrogen. Table 2 lists specifications of gas generators for a 20,000 MT/year production facility. The installed cost of hydrogen production was calculated using the National Renewable Energy Laboratory's H2A model, version 3.0, and the capacity was scaled down from the baseline to the desired production size in this work via the default scaling factor exponent of 0.65.[37] An uninstalled capital cost of $327/kW and a system energy consumption of 50 kWh/kg were used for hydrogen production in our economic analysis.[38] The capital cost of PSA is estimated via the six-tenth's rule[39] based on the base capital cost of $32,539 for 12.5 kg N_2/h production.[40]

Table 2. Specifications of Hydrogen and Nitrogen Generators for 20,000 MT NH_3 Production per Year

Electrolysis		Reference
Technology	Alkaline Electrolyzer	38
Required Hydrogen Output	432 kg/h	
System Energy Efficiency	50 kWh/kg	
Uninstalled Capital Cost	$327/kW	
Delivery Pressure	10 bar	
H_2 Purity	99.999%	
Ambient Temperature	25 °C	
H_2:O_2 by Weight	1:8	
Feed Water	3858 kg/h	
Air Separation Unit		20
Technology	Pressure Swing Adsorption	
Production Rate	2000 kg/h	
N_2 Purity	99.99%	
Power Requirement	0.11 kWh/kg N_2	

Capital Cost. The capacity of the equipment was obtained from simulation and then imported in eq 2 to calculate the purchased equipment costs.

$$\log_{10} C_p^0 = K_1 + K_2 \log_{10}(A) + K_3[\log_{10}(A)]^2 \quad (2)$$

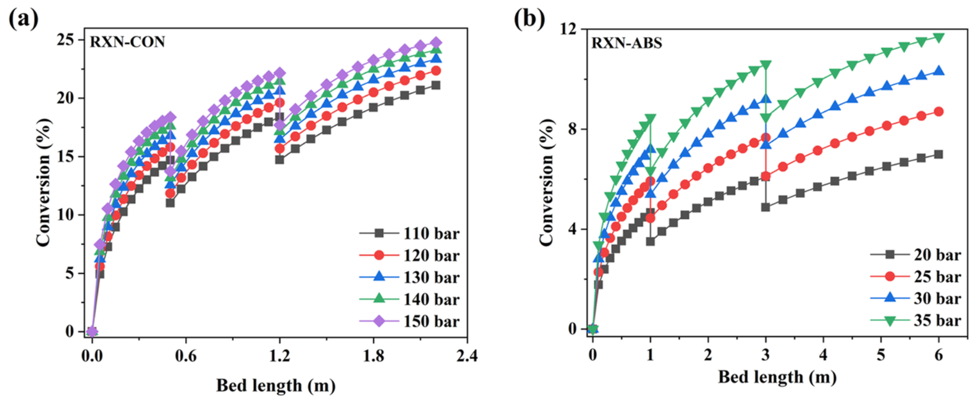

Figure 4. Profile of the conversion along the bed for (**a**) RXN-CON at higher pressure operation (110–150 bar) and (**b**) RXN-ABS at lower pressure operation (20–35 bar).

where K_1, K_2, and K_3 for different equipment are provided by Perry.[36] By incorporating material and pressure factors, the bare module cost is determined using eq 3.

$$C_{BM} = C_P^0\left(\frac{I_2}{I_1}\right)F_{BM} = C_P^0\left(\frac{I_2}{I_1}\right)(B_1 + B_2F_MF_P) \tag{3}$$

where C_{BM} is the bare module cost at nonbase conditions. The I_1 and I_2 are the Chemical Engineering Plant Cost Index for September 2001 and October 2019, respectively.[41] F_M is the material factor, F_P is the pressure factor, and F_{BM} is the bare module factor. B_1 and B_2 are provided by Perry.[36] It is necessary to account for other costs associated with the capital costs, such as contingency, fees, and auxiliary facilities, which lead to the grassroots costs that can be calculated from eq 4:

$$C_{GR} = 1.18\sum_{i=1}^{n} C_{BM,i} + 0.50\sum_{i=1}^{n} C_{BM,i}^0 \tag{4}$$

where C_{GR} is the grassroots costs, and $C^0_{BM,\,i}$ is the bare module cost at base conditions. C_{GR} is also known as the fixed capital investment (FCI), from which the total capital investment (TCI) is calculated as the summation of working capital costs and fixed capital costs, as shown in eq 5.

$$\text{TCI} = \text{FCI} + \text{WC} = \text{FCI} + 0.15\text{FCI} \tag{5}$$

Operating Costs. The cost of manufacturing comprises three main components as shown in eq 6:

$$\text{COM} = 0.180\text{FCI} + 2.73C_{OL} + 1.23(C_{UT} + C_{WT} + C_{RM}) \tag{6}$$

where C_{OL}, C_{UT}, C_{WT}, and C_{RM} are the costs associated with the operating labor, utilities, waste treatment, and raw materials, respectively.

Finally, the levelized cost of ammonia (LCOA), which is defined similarly to the LCOE developed by the U.S. Department of Energy National Renewable Energy Laboratory (NREL), was introduced to evaluate the economics of ammonia production in two processes as given in eq 7:[42]

$$\text{LCOA} = \frac{\text{TCI} + \sum_{t=1}^{n_{life}} \frac{\text{COM}}{(1+r)^t}}{\sum_{t=1}^{n_{life}} \frac{P_t}{(1+r)^t}} \tag{7}$$

where r is the discount rate, P_t is the production rate, and n_{life} is the plant lifetime. The LCOA is defined as the sum of the net present value of the capital and operating costs divided by the overall ammonia production over the lifetime of the project. The LCOA represents the average ammonia price at which an investor would break even for the interest rate of the investment.

RESULTS

In this section, we present the results from the analysis of our process simulation and the economics of the high-pressure RXN-CON versus low-pressure RXN-ABS. Our goal was to understand how the reduced operating pressure can benefit the HB process. More importantly, a lower operating pressure will result in an inherently safer design, considering the fact that such small processes will be operated by nontechnical operators. Operating in moderate conditions is critical to realize the idea of distributed commodity chemical manufacturing. We varied the operating pressure and compared the performance of the high-pressure RXN-CON (110–150 bar) with the low-pressure RXN-ABS (20–35 bar). The comparison of two ammonia processes was carried out in terms of ammonia mole fraction and reactant temperature over a three-bed catalytic converter. It should be noted that process units were sized and designed to achieve the desired ammonia production rate of 20,000 MT/year. We then limited our analysis to RXN-CON operating at 150 bar and RXN-ABS operating at 20 bar and then compared these processes with respect to their energy consumption. Stream tables for the operating condition for RXN-CON (150 bar) and RXN-ABS (20 bar) processes are provided in Tables S5 and S6 in the Supporting Information Section 1.2. This section is then followed by the economic analysis of the proposed configuration illustrated in Figure 2. Details of the capital and operating costs can be seen in the Supporting Information Sections 1.6 and 1.7.

PERFORMANCE COMPARISON

Conversion. The conversion versus reactor length for the RXN-CON and RXN-ABS are illustrated in Figure 4. The converter in the RXN-CON process consists of three beds (0.5, 0.7, and 1 m long), while the RXN-ABS beds are longer by a factor of 2.72 (1, 2, and 3 m long). The RXN-ABS converter is longer because the reaction rate at a lower pressure is slower, and therefore, a larger reactor size is required to achieve the 20,000 MT/year capacity. As the main stream enters R-101, conversion increases as the reaction proceeds. The slope of curves shown in Figure 4 could be considered as the apparent rate of ammonia production. The overall conversion in the RXN-CON process is approximately 25% at 150 bar, which is two times higher than that of RXN-ABS at 35 bar. In both proposed process configurations, the conversion achieved in the first bed is 75% of the overall single-pass conversion. Although the reactor length increases in the subsequent beds, the conversion is only increasing by ∼26% and ∼ 20% of the overall single-pass conversion in the second and the third bed of the convertor R101. This is mainly attributed to the fact that the ammonia partial pressure is growing in the beds; hence, the production rate is more controlled by the reverse reaction. It is important to note that in the RXN-CON process, the gas mixture that enters the converter contains 1.5 mol % of ammonia because the phase-changing condensation is not capable of separating 100% of ammonia (see Figure S2a), whereas the ammonia mole fraction in the gas mixture entering the converter in the RXN-ABS process (Figure S2b) is close to zero because the absorber column almost completely removes the ammonia from the recycle gas until it breaks through.

The ammonia single-pass conversion in the RXN-CON process is higher than the RXN-ABS process, as shown in Figure 4a and b. The kinetics of the ammonia synthesis unanimously suggest that the operating pressure is the major driving force to achieve higher single-pass conversion (see the Supporting Information Section 1.1). The overall conversion in RXN-ABS at 20 bar is 6%, which is about four times lower than that of RXN-CON at 150 bar. In general, increasing the pressure by 10 bar in RXN-CON enhances the conversion by 1% in the R-101 outlet, while conversion can be increased by 3% in RXN-ABS by increasing the pressure by 10 bar. This suggests that the production rate can be more easily improved at the lower pressure and the reverse reaction is controlling at higher ammonia partial pressure.

Space Velocity. Another important factor that significantly affects the single-pass conversion and production rate is the space velocity (inverse of space time) of the gas mixture through the reactor. At lower-pressure processing, it is critical to increase the space velocity to keep the production rate at 20,000 MT of ammonia per year. As displayed in Figure 5, the space velocity in the RXN-ABS process at 20 bar is 8188 h^{-1}, which is 13 times higher than the RXN-CON process at 150 bar. This indicates that the residence time of the gas mixture within the reactor is 13 times shorter; therefore, the ammonia single-pass conversion decreases. Nonetheless, the overall production rate increases because more gas is processed in the synthesis loop at a higher space velocity. Additionally, at a lower pressure, the overall reaction is favored by the forward reaction.

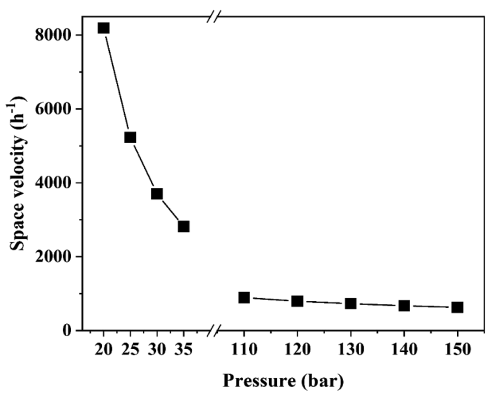

Figure 5. Space velocity through the converter in RXN-CON (110−150 bar) and RXN-ABS (20−35 bar).

Temperature. Controlling the temperature profile of the ammonia converter has been always one of the goals of process optimization efforts.[23,43] The temperature profile for the RXN-CON and RXN-ABS processes versus the reactor length is shown in Figure 6. We assumed an adiabatic reactor; therefore, the temperature of the gas mixture increases as the reaction proceeds in Bed 1. The main stream temperature decreases when mixed with the quenching stream; therefore, the driving force for the reaction increases before the gas mixture enters Bed 2. The major finding from the comparison of RXN-CON and RXN-ABS is that the gas mixture temperature in the converter outlet is approximately 100 °C lower in the RXN-ABS configuration. As discussed above, this is primarily due to low-pressure processing and higher space velocity in the RXN-ABS process (see Figure 5); at a lower pressure, the overall single-pass conversion and the heat of reaction released in the converter is smaller. Hence, the temperature rise in the converter is less severe in the RXN-ABS, which is in favor of more production and smaller reversible reaction rates.

Compression Work. Capital and manufacturing costs associated with the compression are among the major costs in ammonia production. In industrial plants, compressors are the main contributors to energy consumption.[4] The work of compression associated with both feed (C-101) and recycle (C-102) compressors for RXN-CON and RXN-ABS are shown in Figure 7. The work of compression is proportional to the compression factor and the flow rate; in the feed compressor, the work of compression is more influenced by the compression factor, while the work of compression for the recycle compressor is more influenced by the gas flow rate. Results indicate that as the operating pressure increases, the power requirement for the feed compressor increases while that of the recycle compressor decreases on both RXN-CON and RXN-ABS.

For the high-pressure RXN-CON process, the feed compressor should increase the feed pressure from 10 bar to the synthesis loop pressure, which typically requires a chain of compressors with dedicated cooling systems. Since the single-pass conversion and production rate is higher at the high-pressure RXN-CON, there is less need for recirculating an extensive amount of gas in the process. This is consistent with the recycle-to-feed ratios, shown in Figure 7. This ratio for the RXN-CON process is in the range of 3−4. In contrast, RXN-ABS requires much higher recycle compression work. This could be justified by reviewing Figure 5 showing that larger volumes of unreacted hydrogen and nitrogen gases must be recycled in the synthesis loop to achieve the target production rate. Overall, the feed compression work is dominating the energy requirement in the RXN-CON synthesis loop, while the higher work associated with recirculating the unreacted gas in the RXN-ABS process dominates the synthesis loop energy requirement in the RXN-ABS process. It should be noted that the refrigeration compressor also consumes an extra 150 kW for RXN-CON at 150 bar, based on the simulation results.

Heat Duty. Heat exchangers are crucial in the HB process, and the effect of the operating pressure can be easily observed on the extent of the heat exchanging requirements. As shown in Figure 2, E-101 and E-102 are preheaters to warm up the reactants, while E-103 and E-104 are used to cool down the product stream with cooling water and refrigerants, respec-

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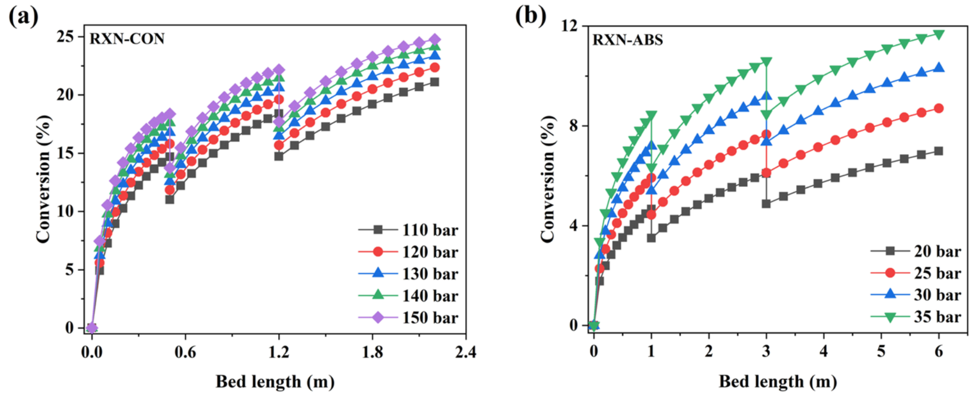

Figure 6. Temperature profile along the bed for (a) RXN-CON at higher pressure and (b) RXN-ABS at lower pressure.

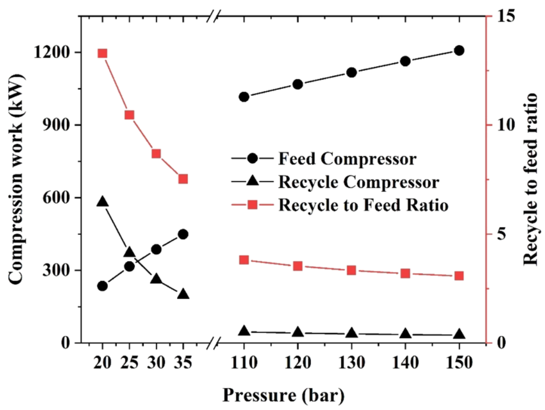

Figure 7. Compression work and recycle ratio for RXN-CON and RXN-ABS.

tively. E-104 is only needed in the RXN-CON process because the product should be chilled to a low enough temperature that ammonia condenses. In the RXN-ABS process, the converter gas outlet will be directed to the absorber column that is in regeneration mode and provides the heat of desorption. Thus, ABS-101 is assumed to have a shell for the heat exchanging, in which heat recovery takes place during the preheating step in the PSA/TSA. Figure 8 shows that the required heat exchanger area for E-101 and E-102 decreases as the pressure increases, which is true for both high-pressure RXN-CON and low-pressure RXN-ABS. These two heat exchangers contribute the most to the heat integration within the process. In RXN-ABS, the residence time of the gas in E-101 and E-102 is less; therefore, more area is needed to efficiently exchange the heat between the reactor feed and the effluent.

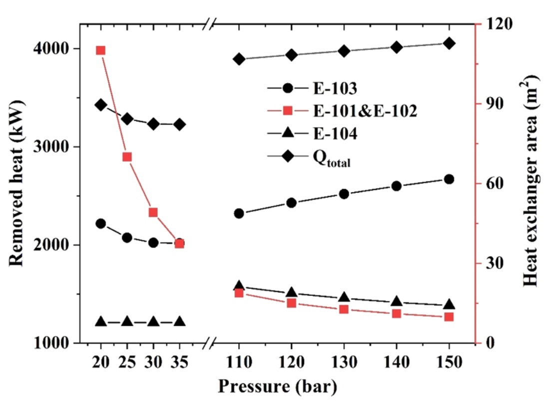

Figure 8. Heat exchanger area and heat removal for RXN-CON and RXN-ABS configurations. E-101 and E-102 are for heat integration of the converter inlet and outlet. Refrigeration is shown as E-104 for RXN-CON. E-103 is water cooled in RXN-ABS. The total heat removed from the synthesis loop is shown as Q_{total}.

After recovering heat, the temperature in the reactor effluent is still relatively high for efficient separation. In RXN-ABS, the water-cooled E-103 seems to be sufficient for cooling the converter effluent stream after heat integration with E-101 and E-102. Less heat is removed in E-103 in RXN-ABS in comparison with RXN-CON. For the RXN-CON process, however, more heat should be removed by the refrigeration system (simplified as E-104) to separate ammonia using phase-changing condensation. As shown in Figure 8, the heat duty for E-104 in RXN-CON decreases as the pressure increases, which indicates that ammonia is more easily condensed because of lower condensation temperature and longer residence time at higher partial pressure.

Absorber Specifications. After solving the mathematical model, breakthrough time with three temperatures, 150, 200, and 300 °C were estimated. The model can accurately predict the breakthrough time at different temperatures with less than 1.4% error compared to the experimental results.[18] Details of the accuracy of the model validation are given in Table S7 in the Supporting Information Section 1.4. Under the assumption that the scaled-up model is valid for RXN-ABS ammonia production of 2428.8 kg/h, the absorber was designed and the specifications are summarized in Table 3. Since the goal of this research was only to investigate the techno-economic analysis of ammonia production with RXN-CON and RXN-ABS, a full-

Table 3. Absorber Design and Specifications[22]

Description	Symbol	Value
Absorber		
Length	L	4.7 m
Diameter	D	1.88 m
Number of Nodes	-	40
Bed Voidage	ε_i	0.32
Absorbent ($MgBr_2$−Si)		
Particle Voidage	ε_b	0.4
Absorbent Density	ρ_s	3720 kg/m^3
Particle Radius	r_p	2×10^{-4} m
Pore Radius	R_{pore}	2e−10 m

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scale dynamic absorber model was not necessary. In the future, we will investigate the absorption and release mechanisms in detail and develop a multibed ammonia absorption unit.

Energy Analysis. Conventional ammonia production is energy-intensive due to (1) the large amount of compression work required in the synthesis loop and (2) energy-intensive phase-changing separation. These two features are interrelated; high-pressure processing is needed for efficient separation (Sherwood Concept). Addressing these two challenges for small-scale processing was the motivation of this work. To reduce the work of compression, the synthesis loop operating pressure was reduced by 87%. At lower pressure, however, the single-pass conversion is not enough high for viable phase-changing separation. It is estimated that exploiting separations that can bypass the phase-changing separations can lead to energy savings of up to one order of magnitude.[44] To improve the efficiency of the separation, the condensation was replaced with high-temperature absorption. Based on the process design and simulation results obtained from ASPEN Plus, the energy consumption for each process configuration was estimated in terms of power requirements. Figure 9 shows the comparative energy analysis of the proposed processes consumption in both RXN-CON and RXN-ABS configurations. As illustrated, energy consumption comprises three main subsystems: hydrogen production, nitrogen production, and ammonia synthesis loop. Since both the processes operate with similar feed, energy consumption for hydrogen and nitrogen production are equal. It is evident that more than 90% of electricity is consumed in hydrogen production.

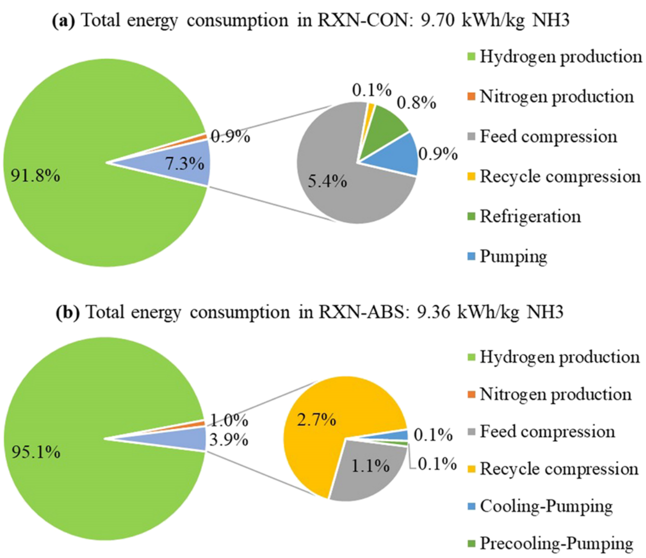

Figure 9. Energy consumption in (a) RXN-CON (150 bar) and (b) RXN-ABS (20 bar). The bigger pie represents energy consumption of the whole plant consisting of hydrogen/nitrogen production and the ammonia synthesis loop. The smaller pie illustrates the energy consumption of unit operations in the ammonia synthesis loop.

Detailed energy consumption in the synthesis loop for RXN-CON and RXN-ABS is summarized in Table 4. The feed and recycle compression are dominating the energy consumption in the RXN-CON and RXN-ABS synthesis loops, respectively. The feed compressor in RXN-CON consumes 74% of the electricity. This is attributed to the increase in the synthesis loop pressure, as illustrated in Figure 7. In the RXN-ABS proposed process. It is important to note that we simulated the refrigeration cycle as a simple Carnot cycle. We presume that the refrigeration will account for higher energy consumption when simulated with all details that is practiced in the industrial scale. There is also more opportunity to optimize the PSA process for cycling the absorber column without using energy-intensive TSA. Nevertheless, results suggest that the operating pressure reduction has a significant effect on the energy requirements of the synthesis loop. If the energy input for the feed production is disregarded, the energy consumption in the RXN-ABS synthesis loop decreases by roughly 50% when the operating pressure is reduced.

Table 4. Energy Consumption in the Ammonia Synthesis Loop for the RXN-CON (150 bar) and RXN-ABS (20 bar)

	Energy Consumption [kWh/kg NH_3]			
Component	RXN-CON	%	RXN-ABS	%
Feed Compression	0.52	74%	0.10	27%
Recycle Compression	0.01	2%	0.25	68%
CW Pumping	0.09	12%	0.01	3%
Refrigeration/Precooling	0.08	12%	0.01	1%
Total	0.70	100%	0.37	100%

Economic Analysis. In this section, capital and operating costs of proposed process configurations are estimated based on the process simulation results and cost estimation. The LCOA for both RXN-CON and RXN-ABS are presented to evaluate the profitability of small-scale, green ammonia production.

Capital Costs. Simulation results were employed to size the equipment and then calculate the bare module costs, as shown in Figure 10. The total bare module cost of RXN-ABS is about $1.02 million less than that of RXN-CON. Among all subsystems, the hydrogen generator is the largest contributor to the capital costs, accounting for more than 60% of the bare module costs. The second major cost in RXN-CON is the cost of compressors ($3.47 million), which is $0.61 million higher than that of RXN-ABS. The cost of heat exchangers in RXN-ABS is slightly less than that in RXN-CON because the low-pressure process does not generate a significant amount of excess heat, thereby less heat exchanger area is needed. Total capital investment was calculated based on the bare module costs as described in eqs 3−5. Overall, the total capital investment of RXN-ABS is $2.65 million less than the capital cost of RXN-CON (see Table S9 in the Supporting Information Section 1.6).

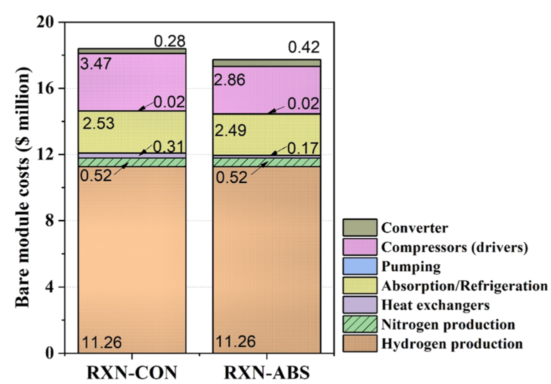

Figure 10. Bare module cost of selected unit operations in RXN-CON and RXN-ABS

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Operating Costs. As discussed earlier, we assumed that our small-scale Haber plant uses wind-generated electricity at a fixed price and grid electricity will be purchased at the time of intermittency, to ensure continuous operation of the plant. Estimating an accurate wind+grid LCOE is rather complicated and needs its own dedicated study, which is beyond the scope of this work. Overall, electricity pricing is a function of various variables, and there is a large variation in the pricing reported in the literature. The current U.S. DOE's Wind Energy Technologies Office estimate for wind-generated electricity is \$10−20/MWh.[45,46] The Lazard wind LCOE estimations are provided in a wider range of \$28−54/MWh.[47] A report by the U.S. DOE Fuel Cell Technologies Office suggested the electricity price of \$37/MWh as a projected cost of electricity for hydrogen production in 2020.[47] A recent U.S. DOE report for the wind market analysis reported the wind LCOE at an all-time low of \$36/MWh.[48] We took the data regarding the availability of wind in our GLEAMM microgrid facility (see Figure S6 in the Supporting Information) as the model for our economic analysis. We hypothesized that the wind capacity factors above 15% are required to carry out the continuous operation of a chemical plant. Results indicated that the wind electricity generation was not sufficient in 17.7% times to ensure continuous production; hence, it was necessary to purchase the electricity from the grid at an average cost of \$64.8/MWh.[49] Therefore, we estimated the average LCOE for combined wind+grid to be \$41.11/MWh.

Operating costs were calculated based on the power requirements for each unit and consumption of the cooling water, refrigerant, iron catalyst, and absorbent. Operating costs consist of three major categories: direct, fixed, and general manufacturing costs. These costs were estimated using eq 6 in terms of five main components, including fixed capital investment, cost of operating labor, cost of utility, cost of waste treatment, and cost of raw materials. The results of operating costs for RXN-CON and RXN-ABS are summarized in Table 5. The major contributor to the operating cost is the direct manufacturing costs that are mostly affected by the electric utility for feed production. The reduction in the synthesis loop pressure and replacement of phase-changing separation led to 9.68% improvement (\$1.67 million) in the operating costs of RXN-ABS.

Table 5. Operating Costs for RXN-CON (150 bar) and RXN-ABS (20 bar)

	Operating Costs [\$Million]			
Subsystem	RXN-CON	%	RXN-ABS	%
Direct Manufacturing Costs	13.80	79.86	12.23	78.47
Fixed Manufacturing Costs	2.94	17.03	2.84	18.23
General Manufacturing Costs	0.54	3.11	0.52	3.30
Total	17.26	100	15.59	100

As summarized in Table 6, the capital costs for RXN-CON and RXN-ABS are calculated to be \$34.26 million and \$32.11 million, with the operating costs of \$17.26 million and \$15.59 million, respectively. In this work, the projected life of the plant (n_{life}) is assumed to be 20 years with the discount rate (r) of 7%.[19] Using eq 7, the on-site LCOA is estimated to be \$1030/MT NH_3 and \$933/MT NH_3 for RXN-CON and RXN-ABS, respectively. The lower LCOA in RXN-ABS is mostly attributed to the savings by lowering the operating and capital costs by reducing the pressure and replacing phase-changing separation. Nevertheless, the projected LCOA for small-scale, all-electric ammonia is about twice more expensive than the conventional ammonia (\$500/MT of ammonia, retrieved from[50]). Such a drastic increase in costs is largely due to energy- and capital-intensive hydrogen production with alkaline electrolyzers, as well as the process capacity reduction.

Table 6. Executive Summary for LCOA with RXN-CON and RXN-ABS

Cost\Process	Unit	RXN-CON	RXN-ABS	% Reduction
Cost of Manufacturing	[\$ Million]	17.26	15.59	9.68
Capital Cost	[\$ Million]	34.26	32.11	6.10
LCOA	[\$/MT]	1030	933	9.42

The large amount of oxygen produced through water electrolysis can be considered as a resource for the all-electric ammonia production. Oxygen has several important applications, such as combustion, semiconductor manufacturing, and wastewater treatment.[50] The economics of small-scale ammonia production can be largely improved by taking into account the revenue gained by selling high-purity oxygen produced, although it poses potential safety risks. In a very rough estimation, we assumed that the oxygen produced could be sold locally, so there will be insignificant expenses on transportation and distribution. By incorporating the cost of the storage tank for oxygen and the resulting revenue from selling the oxygen (no cost of compression and treatment was considered), the net profit is estimated to be \$16.3 million/year (see details in the Supporting Information Section 1.11). A cash flow table (Table S15 for RXN-CON and Table S16 for RXN-ABS in the Supporting Information Section 1.11) was constructed assuming a 20-year project life span. The combined revenue from selling oxygen at \$125/tank[51] and ammonia at the median commodity price of \$512−683/MT (obtained from ref[28] in Jan 2019 to April 2020) has improved the overall economics of the small-scale Haber process, with results showing that both RXN-CON and RXN-ABS ammonia plants are profitable with an after-tax rate of return of 21.83 and 27.50%, respectively. The revenue obtained from selling oxygen was based on a very rough operating and capital cost estimate. A more rigorous model and cost analysis is required for a more accurate estimation.

DISCUSSION

Energy Consumption. To better evaluate the energy consumption in RXN-CON and RXN-ABS processes, large-scale SMR-based RXN-CON ammonia production was selected as our basis, where 1.28 kWh/kg NH_3 is expended in the ammonia synthesis loop.[4] The total energy consumption of 32 GJ/MT NH_3 (8.89 kWh/kg NH_3) was also considered for SMR-based RXN-CON.[4] The energy consumption of the synthesis loop in different forms of ammonia production is summarized in Table 7. Electrolysis-based ammonia production consumes more energy than the SMR-based process in terms of gross caloric value (1 kWh/kg NH_3) which is due to the hydrogen generation energy requirement. Energy consumption in the synthesis loop, however, is smaller than the large SMR-based process, approximately 0.71 and 0.37 kWh/kg NH_3 lower for small-E-RXN-CON and small-E-RXN-ABS, respectively.

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Table 7. Total and Synthesis Loop Energy Consumption in Different Process Configurations

System	Total Energy Consumption (kWh/kg NH_3)	Overall Energy Efficiency(%)	Energy Consumption in Ammonia Synthesis Loop (kWh/kg NH_3)
Small-E-RXN-CON[a]	9.70	67%	0.71
Small-E-RXN-ABS[a]	9.36	67%	0.37
Large-SMR-RXN-CON	8.89	78%	1.28

[a]"E" stands for electrolysis.

Figure 11 illustrates an overview of the energy consumption from different reports using two ammonia production technologies. For RXN-ABS, data were adopted from Rouwenhorst et al.[28] and Palys et al. (the University of Minnesota's Morris pilot plant),[22] while for RXN-CON, data were adopted from Reese et al. (Morris plant),[10] Rouwenhorst et al. (Proton Venture NFUEL and ThyssenKrupp),[28] Morgan et al.,[52] Pfromm,[8] and Brown (large-scale alkaline).[53] Fitted curves for RXN-CON and RXN-ABS indicate that energy consumption for all-electric ammonia production can be significantly reduced with the increasing production size. At larger scales, energy consumption by RXN-ABS tends to coincide with that of RXN-CON, which is most likely to happen because hydrogen production consumes the majority of energy for these two electrolysis-based ammonia processes. Energy consumption slowly drops in the vicinity of 9 kWh/kg NH_3 at 1000 MT/year production, which is approximately identical to that of the large-scale RXN-CON production. However, as small-scale ammonia production is becoming promising, low-pressure RXN-ABS will outperform high-pressure RXN-CON in terms of energy consumption. Moreover, the energy efficiency of alkaline electrolyzers is expected to be improved in the future.[38] As a result, this will further reduce energy consumption for renewable ammonia production.

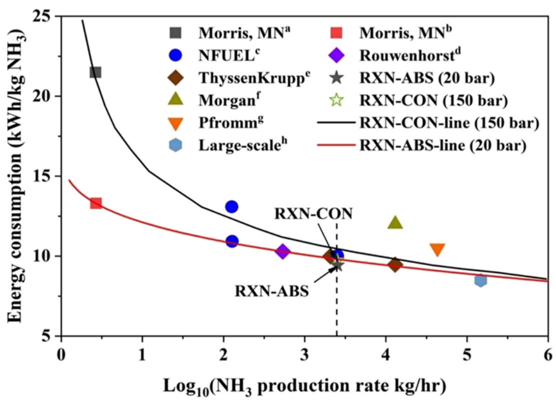

Figure 11. Energy consumption for electrolysis-based ammonia production. Data were adopted from the following. RXN-CON production: subscript a is for Morris, MN,[10] c for Proton Venture NFUEL,[28] e for ThyssenKrupp,[28] f for Morgan et al.,[52] g for Pfromm,[8] and h for large-scale alkaline;[53] and RXN-ABS: subscript b is for Morris, MN[22] and d adopted from Rouwenhorst et al.[28]

Sensitivity of LCOA to LCOE. The results presented indicate that small-scale all-electric ammonia production is still energy-intensive, compared with the conventional ammonia. Energy consumption is mainly dominated by electricity-driven hydrogen production, and the LCOE has a direct impact on the LCOA. According to Lazard's report, the marginal price of wind-generated electricity is \$28−\$54/MWh without federal subsidies, but it can further decrease to \$11−\$45/MWh considering subsidies.[11] In addition, renewable electricity penetration is increasing rapidly. It is projected that locational marginal pricing (LMP) for wholesale electricity in California and Texas in 2030 significantly drops and might lead to low-to-negative LMP during the day.[54,55] In a high variable renewable energy penetration scenario with variable grid conditions, it is expected to have negative LMP in the near future, in areas with high wind speed or solar radiation.[56] Hence, the question that may arise is whether or not the intensified all-electric ammonia Haber process could be used as a means of thermochemical energy storage for the scenario that renewable-based electricity is becoming cheaper and subsidized.

Figure 12 shows the sensitivity of the LCOA to LCOE varying from \$0 to \$54/MWh in RXN-CON and RXN-ABS

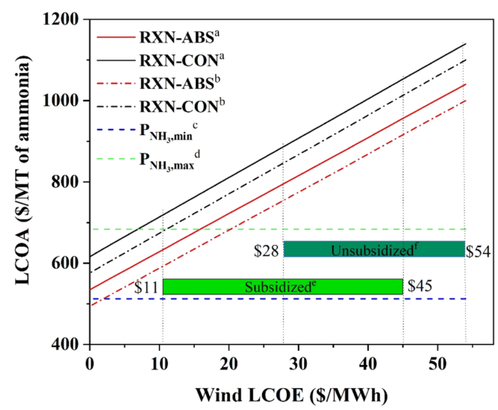

Figure 12. LCOA sensitivity to wind LCOE. Superscript a denotes the LCOA for RXN-CON (black solid line) and RXN-ABS (red solid line) with the current electrolyzers (\$327/kW and 50 kW/kg); Superscript b denotes the LCOA for RXN-CON (black dotted line) and RXN-ABS (red dotted line) with the future electrolyzers (\$215/kW and 46 kW/kg); Superscripts c and d are commodity price of ammonia in Jan 2019 (green dashed line) and April 2020 (blue dashed line). Data source: USDA AMS;[49] subscripts e (light green region) and f (dark green region) indicate the range of subsidized and unsubsidized wind LCOE.[11]

processes, where electrolyzers with different energy efficiencies and uninstalled costs are evaluated. The target of electricity usage and uninstalled cost of alkaline electrolysis in 2020 are set at 46 kWh/kg of hydrogen and \$242/kW, respectively.[38] It is obvious that the LCOA increases proportionally with the LCOE. As the LCOE approached the lower-bound value of \$11MWh, the LCOA in RXN-ABS fell in the commodity price range of \$512−683/MT of ammonia during Jan 2019 to April 2020.[49] It is important to note that the conventional commodity ammonia processing takes advantage of economy of scale and does not include the cost of transportation. The prices reported for conventional ammonia do not include the scenario in which the carbon taxes might be imposed that can significantly decrease the profitability of ammonia produced from SMR. By accounting for the lower-bound price of

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subsidized wind LCOE via the current electrolyzers, the LCOA in RXN-CON and RXN-ABS can be cut down to \$720/MT and \$680/MT of ammonia, respectively, while it can be further reduced to \$634/MT and \$594/MT of ammonia with the future electrolyzers, respectively. In some regions with negative LMP, the LCOA can be even reduced to \$494/MT of ammonia, which is even lower than the commodity price based on large-scale SMR-based RXN-CON.

Learning Rates for Modular Manufacturing. Compared to the large-scale conventional process, all-electric ammonia production is more expensive primarily due to the smaller production capacity and cost of hydrogen.[57] The capital cost for commodity chemicals usually follows the six-tenth's rule; therefore, making green ammonia in a small scale is inherently less favorable. The RXN-ABS configuration can be further intensified in modularized units and be used in sustainable farms or for thermochemical energy storage.[58] With the learning rate obtained from modular ammonia production, there is a great chance to further reduce costs in terms of LCOA based on the experience and knowledge obtained from manufacturing more modular units. The learning curve was introduced to predict the possible reduction in the capital investment for modularized ammonia production, as defined in eq 8:[59]

$$\mathrm{TCI}_M = \mathrm{TCI}_1 M^{-b} \tag{8}$$

where TCI_1 is the total capital investment to produce the first modular ammonia plant, and M is the cumulative number of modular ammonia plants. TCI_m is the total capital investment to produce the M_{th} modular ammonia plant by means of the learning parameter, and b represents the rate of cost reduction.[59] The learning rate, LR, is defined by eq 9, which denotes the cost reduction associated with doubling cumulative modular unit production (or capacity).

$$LR = 1 - 2^{-b} \tag{9}$$

The learning rate of 20% was adopted to calculate TCI_M for different numbers of modules,[60] and the resulting TCI_M was normalized by the corresponding values of M. With the learning rate of 20%, the total capital investment can be reduced by about 94% for both RXN-CON and RXN-ABS with the cumulative production of 5000 modules. As a result, the LCOA for RXN-CON decreased from \$1030 to \$889/MT of ammonia, while that of RXN-ABS decreased from \$933 to \$798/MT of ammonia, as displayed in Figure 13.

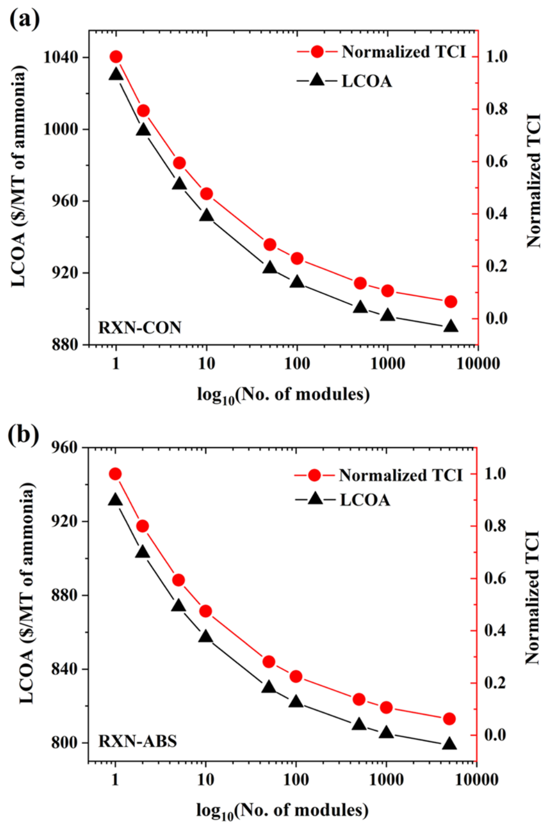

Figure 13. Comparison of learning curves for (a) RXN-CON and (b) RXN-ABS; electricity price: \$41/MWh; electrolyzer: \$327/kW and 50 kW/kg.

CONCLUSIONS

In this work, we investigated the techno-economic analysis of an small-scale, all-electric ammonia production process from totally renewable resources. Condensation, commonly used in the HB process, was replaced by absorption to separate ammonia, which leads to reducing the operating pressure in the synthesis loop. With varying operating pressures, the performance of RXN-CON (110−150 bar) and RXN-ABS (20−35 bar) were evaluated with regard to the reaction conversion, temperature, compression work, and heat duty. Analysis of operating and capital costs shows that the uninstalled cost of hydrogen electrolyzer, cost of hydrogen production, and cost of compression are dominating the small-scale, all-electric ammonia production. The results of the economic analysis indicate that by replacing condensation with absorption in a 20,000 MT per year process, capital costs and annual operating costs are reduced by 7.73 and 9.68%, respectively, and the LCOA decreases by 9.61% when low-pressure RXN-ABS is used.

The LCOA was further optimized for different LCOE in different scenarios. Results indicate that all-electric ammonia is around two times more expensive than the conventional commodity ammonia. The production based on RXN-ABS falls within the cost-competitive range when the LCOE to this plant drops below \$10/MWh. When there is a potential for negative LMP at a high renewable penetration scenario, the resulting LCOA in RXN-ABS was lower than the commodity price of ammonia, which will make the RXN-ABS process profitable. As the research and development on the energy efficiency and costs of electrolyzers continue, cost-effective and energy-efficient electrolyzers will further improve the LCOA. Moreover, the empirical data and knowledge obtained from small-scale ammonia manufacturing will be advantageous for further reduction in the unit capital investment. Small, modular processes can provide a pathway to replace the economy of scale with the economy of numbers.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acssuschemeng.0c04313.

Ammonia synthesis kinetics embedded in the reactor of RXN-CON and RXN-ABS; values for A, B, and C for evaluating the activities for synthesis kinetics; equilibrium conversion versus temperature; Design specifications for RXN-CON and RXN-ABS processes; ammonia mole fraction along the bed in RXN-CON and RXN-ABS; converting ammonia mole fraction to conversion based on simulation results; Stream table for the operating condition for RXN-CON and RXN-ABS processes; summary of the equations of the dynamic absorption column model; Breakthrough time of ammonia absorption; Ammonia capacity of MgBr2−Si at different temperatures; capital costs for RXN-CON and RXN-ABS; operating costs for RXN-CON and RXN-ABS; wind generation capacity; and nondiscounted cumulative cash flow. (PDF)

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Notes

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NOMENCLATURE

Abbreviations

CEPCI:Chemical Engineering Plant Cost Index
HB:Haber−Bosch
LCOA:Levelized cost of ammonia
LCOE:Levelized cost of energy
LMP:Locational marginal pricing
PSA:Pressure swing adsorption
PSA/TSA:Pressure and temperature swing absorption
RXN-ABS:Reaction-absorption
RXN-CON:Reaction-condensation
SMR:Steam methane reforming

ROMAN SYMBOLS

b:Rate of cost reduction
C_{BM}:Bare module cost [$ million]
C_{GR}:Grass-roots cost [$ million]
COM:Cost of manufacturing [$ million]
C_p^0:Purchased equipment cost [$]
D:Absorber column diameter [m]
FCI:Fixed capital cost [$ million]
I:Chemical Engineering Plant Cost Index
L:Absorber column length [m]
LR:Learning rate
M:Cumulative number of modular ammonia plants
n_{life}:Plant lifetime (year)
P_t:Production rate (metric ton/year)
r:Discount rate
r_pg:Particle radius (m)
R_{pore}:Pore radius (m)
TCI:Total capital investment ($ million)
WC:Working capital cost ($ million)

GREEK SYMBOLS

ε_i:Interparticle voidage
ε_p:Intraparticle voidage
ρ_g:Gas density [kg/m^3]
ρ_s:Absorbent density [kg/m^3]

SUBSCRIPTS AND SUPERSCRIPTS

BM:Bare module
g:Gas
GR:Gross root
i:Interparticle
0:Denotes the base condition for equipment cost
p:Particle
$pore$:Absorbent particle pore
s:Solid (refers to absorber particle)

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